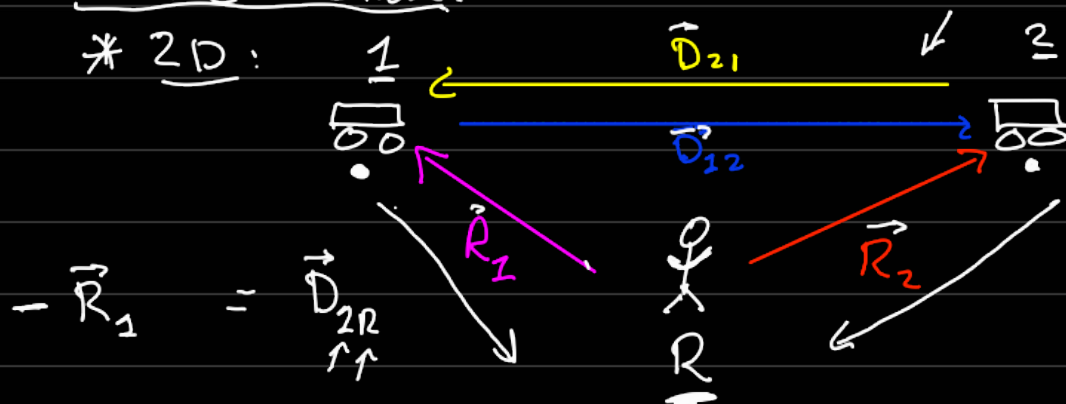


* Circular Motion: $\vec{r} = x(t)\hat{e}_x + y(t)\hat{e}_y + z(t)\hat{e}_z$
 $= r\hat{e}_r$
 $= r(\cos(\theta)\hat{e}_x + \sin(\theta)\hat{e}_y)$

• Polar coordinates: coord: different bases ($\hat{e}_r, \hat{e}_\phi$)

• Relative motion:



$\vec{D}_{12}$: Dist. measured by driver 1 between him and car 2
 $\vec{D}_{12} = -\vec{D}_{21}$

• obj are moving very fast.

• Add: "Head-to-tail": $\vec{R}_1 + \vec{D}_{12} = \vec{R}_2$ *

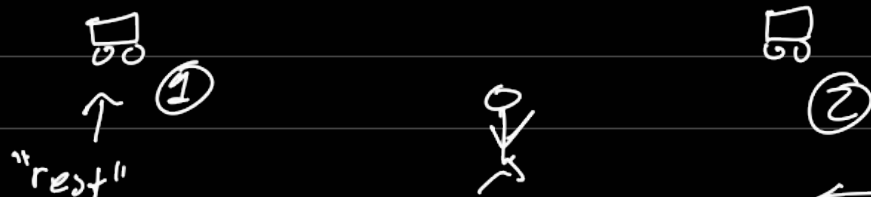
• Velocities: $\frac{d\vec{D}_{12}}{dt} = \vec{V}_{12}$
 $\frac{d\vec{R}_1}{dt} = \vec{V}_1, \quad \frac{d\vec{R}_2}{dt} = \vec{V}_2$

$$\vec{V}_1 + \vec{V}_{12} = \vec{V}_2 \quad | \quad \vec{V}_1 = \vec{V}_{21} + \vec{V}_2$$

$$\vec{V}_{21} + \vec{V}_2 + \vec{V}_{12} = \vec{V}_2 \quad \Leftrightarrow \quad \vec{V}_{12} = -\vec{V}_{21}$$

$\xrightarrow{50 \text{ mph}} \quad \quad \quad \xleftarrow{70 \text{ mph}}$

ex:



To car ①: car ② looks like its going
 - 70 mph

4. car ②: car ① is moving at 70 mph

To R: (1) +50 mph
(2) -20 mph

* Moving fast: Distance is not nec. invariant

"distance": $-(c\Delta t)^2 + x^2 + y^2 + z^2$

(Aside) is

c: speed of light

$\Delta x = v\Delta t$, $v = \frac{\Delta x}{\Delta t}$

$\Delta x = c\Delta t$

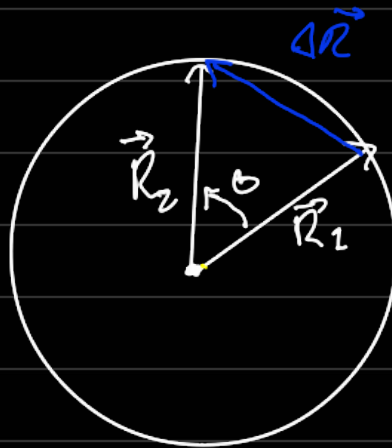
(later)

"Normal distance"

Circular motion:

Vel: $V = \frac{\Delta x}{\Delta t}$ 1D

3D: $\vec{V} = \frac{d\vec{R}}{dt}$



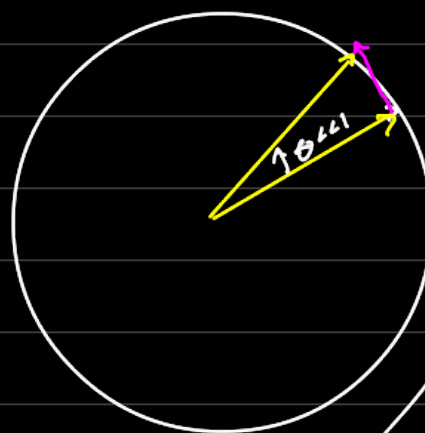
$\vec{R}_1 + \Delta\vec{R} = \vec{R}_2 \rightarrow \Delta\vec{R} = \vec{R}_2 - \vec{R}_1$

* $\vec{V} = \frac{\Delta\vec{R}}{\Delta t}$ (tells us that $\vec{V}$ points along $\Delta\vec{R}$)

$\lim \Delta t \rightarrow 0: \theta \rightarrow 0$

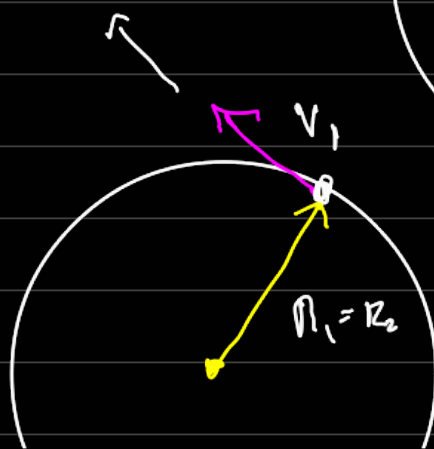
• Know $\vec{V}$ looks like

$\theta \rightarrow 0$



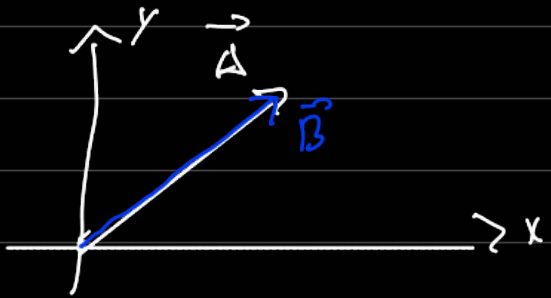
$\vec{V}$: tangent to $\vec{R}$
 $\vec{V} \perp \vec{R}$

but that
 $\theta = 0$



* Vel. tangent/perp to $\vec{R}$
at all times

ex: $\vec{A} = A_x \hat{e}_x + A_y \hat{e}_y$
 $\vec{B} = B_x \hat{e}_x + B_y \hat{e}_y$



$$\vec{A} = \vec{B}$$

$$A_x \hat{e}_x + A_y \hat{e}_y = B_x \hat{e}_x + B_y \hat{e}_y$$

$$(A_x - B_x) \hat{e}_x + (A_y - B_y) \hat{e}_y = 0$$

$$A_x = B_x$$

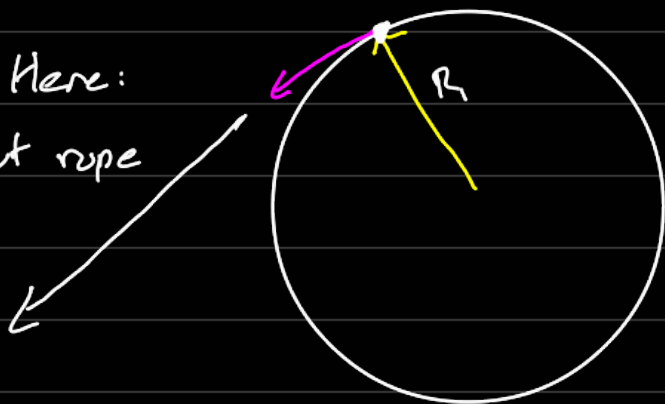
$$A_y = B_y$$

ex: $\frac{df}{dx} = \lim_{\Delta x \rightarrow 0} \left(\frac{f(x + \Delta x) - f(x)}{\Delta x} \right)$

• at a later time

* Here:

cut rope



* Result: Newton's law: $\sum \vec{F} = m\vec{a}$

→ acceleration:

1st ex: $\vec{a} = 0$: $F = 0$ (no force)

$$\vec{a} = \frac{\Delta \vec{V}}{\Delta t} \quad \Bigg| \quad \vec{V} = \frac{\Delta \vec{R}}{\Delta t}$$

• look at geometry

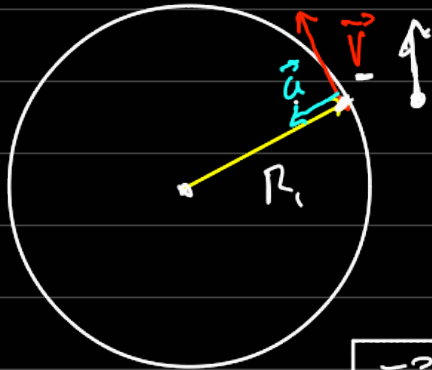




$$V_1 + \Delta V = V_2$$

$$\Delta \vec{V} = \vec{V}_2 - \vec{V}_1$$

$\vec{a} = \frac{\Delta \vec{V}}{\Delta t}$: Points towards the center of circle
 $\theta \rightarrow 0$



$\vec{a}_c$: Centripetal acceleration
 for uniform constant motion

$$\boxed{\vec{F}_c = m \vec{a}_c} \quad *$$

* Reference frames:

Newton $\vec{F} = m\vec{a}$

Here: we find an acc. $\vec{a}_c$ and we call
 the cent. force

$$\vec{F}_c = m \vec{a}_c \leftarrow$$

R_1 : position vector of Ball

$\vec{V}$: velocity vector of Ball

$\vec{a}_c$: acc. vector of Ball

Real! Force: String in this / $\vec{a}_c$

* Person on circle: feels 'fake' force

